

Statistical Machine Learning Project

Project Goal

The purpose of this project is to give students hands-on experience applying Statistical Machine Learning methods to a real dataset while exploring tools beyond what was explicitly covered in class.

Students will:

- select an interesting dataset,
- choose **either R or Python** as their primary implementation language,
- focus on one project component,
- implement **at least one method or approach that was not taught in class**.

The emphasis is on exploration, justification, and clear comparison.

1. Dataset Selection

Choose a dataset that is:

- interesting and meaningful,
- large enough to require modeling decisions,
- suitable for machine learning analysis.

Possible sources include:

- UCI Machine Learning Repository
- Kaggle
- OpenML
- Government or public data portals
- Domain-specific datasets (health, finance, environment, etc.)

You must briefly explain:

- why the dataset is interesting,
- the prediction or analysis goal,
- key variables and challenges.

2. Choose One Implementation Track

You must select **one language**:

- **R track**, or
- **Python track**.

Below is a side-by-side overview of common components.

3. Side-by-Side Component Options

Project Component	R (examples)	Python (examples)
Data wrangling	<code>dplyr</code> , <code>tidyr</code> , <code>data.table</code>	<code>pandas</code> , <code>numpy</code> , <code>polars</code>
Visualization	<code>ggplot2</code> , <code>plotly</code>	<code>matplotlib</code> , <code>seaborn</code> , <code>plotly</code>
Classical ML models	<code>tidymodels</code> , <code>caret</code> , <code>glmnet</code>	<code>scikit-learn</code> , <code>xgboost</code>
Regularization & feature selection	<code>glmnet</code> , <code>grpreg</code>	<code>sklearn.linear_model</code> , <code>feature_selection</code>
Functional / advanced modeling	<code>fda</code> , <code>refund</code> , <code>mgcv</code>	<code>scikit-fda</code> , custom implementations
Neural networks / deep learning	<code>torch</code> , <code>keras</code>	PyTorch, TensorFlow/Keras
Model evaluation	<code>yardstick</code> , <code>rsample</code>	<code>sklearn.metrics</code> , <code>cross_val_score</code>
Reporting / reproducibility	Quarto / R Markdown	Jupyter / Quarto

4. Required Novel Component

You must implement **at least one approach not explicitly taught in class**.

Examples include:

- a new model class,
- a different regularization strategy,
- advanced cross-validation or tuning,
- feature engineering approach,
- functional or structured data method,
- neural network variation,
- explainable ML method.

Your report must include:

- why you selected this approach,
- how it differs from class material,
- what you learned from using it.

5. Minimum Project Structure

Your final submission should include:

(a) Introduction

- Problem description
- Dataset overview
- Research question

(b) Data Exploration

- Summary statistics
- Visualizations
- Preprocessing decisions

(c) Modeling

- Baseline model(s)
- New (unseen) method
- Training procedure

(d) Evaluation

- Performance metrics
- Comparison of methods
- Interpretation of results

(e) Discussion

- Strengths and limitations
- What worked / what failed
- Future improvements